1. Innovation

Problem Type 1 Diabetes (T1D) patients take multiple daily insulin doses but lack tools to translate data into safe, personalized guidance. Poor management causes dangerous hypoglycemia and long-term complications [1]. A "precision gap" exists: doctor-set therapy plans are difficult to apply correctly daily. Most people with Type 1 Diabetes use CGM or finger-prick tests that show glucose levels but do not help decide insulin doses, leading to guesswork, frequent corrections, and high mental stress. [2]

Technology DiaTwin is a clinically supervised decision-support platform delivering pump-like precision for standard insulin pens. Physicians preset all parameters (insulin-to-carb ratios, correction factors, basal rates, safety limits), which are validated and remotely monitored. The platform combines pharmacokinetic modelling with machine learning to provide safe, personalized dosing recommendations.

Dual-Target Approach:

- **CGM users (Germany/EU)**: Translates glucose trends into actionable advice via Net Effect analysis, 4-hour projection, and insulin-on-board (IOB) tracking
- **Pen users (global)**: Uses 3-4 daily finger-sticks with Intelligent Glucose Estimation and Meal-on-Board (MOB) modelling

Core features: Dual Net Effect System (physician validation + patient projection); Intelligent Glucose Estimation (physiology-based interpolation); MOB (5 meal types, easy portion inputs); Future Glucose Projection (IOB + MOB).

Current Readiness (TRL 4-5): Core algorithms validated through Master's thesis (grade 1.6 "Excellent") with 50,000+ meal-outcome pairs from real clinical data. Clinical Utility Score: 8.7/10 from certified diabetologist. Web + mobile prototypes operational.

Next Phase (TRL 6-7): Prospective clinical trial (N=100 patients, 6 months) for safety/efficacy validation, physician workflow integration, and regulatory compliance (CE Class IIa + DiGA pathway).

2. Market situation

Market Volume: Germany has ~1.3 million insulin-pen users with CGM/finger-prick testing who lack structured dosing guidance, DiaTwin's primary target. Globally, insulin-pen therapy dominates due to pump/CGM costs and limited availability.

Market Growth: Global diabetes device and digital management markets growing at 8.7% CAGR [3]. Software-based decision support represents one of the fastest-growing segments.

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Market Potential: Conservative 2% penetration = ~26,000 German users within 3 years. DiGA reimbursement pathway creates sustainable revenue model with strong international expansion potential where hardware access is limited [4].

3. Unique selling proposition and customer benefits

DiaTwin delivers CGM-level precision and intelligent insulin dosing guidance without continuous sensor costs, acting as a CGM Companion & Sustainability Tool. For CGM users, it adds structured dosing guidance to trend awareness, reducing unnecessary corrective doses and cognitive burden. For insulin-pen users, it provides trend insight and automated meal-based dosing recommendations**.**

Feature	DiaTwin	CGM-Only	Basic Apps	Insulin Pumps
Intelligent Dosing Guidance	YES (IOB, MOB, Projection)	NO	NO	YES (Closed-loop)
Cost Reduction	95% (€100/year)	NO (€3k-5k/year)	N/A	NO (€10K-16K)
With Insulin Pens	YES	YES	YES	NO

DiaTwin delivers significant benefits: **Post-CGM Users** save €3,000-5,000/year while maintaining clinical precision. **Continuous CGM Users** gain analytical enhancement and parameter auto-optimization. **Physicians** increase efficiency by reducing parameter validation time from months to weeks and gain transparency via remote monitoring**. Healthcare Systems** realize substantial cost savings (€10,500-27,000 per patient/year in avoided ER visits) and promote health equity.

4. Know-how support

Team Lead: Ali Attia – Pharmacist, M.Sc. Digital Health (DIT); 5+ years Digital Transformation Officer in healthcare.

Academic Foundation: Master's thesis at DIT (grade 1.6 "Excellent"), Core algorithms (pharmacokinetic modeling per S3 Guidelines, dual Net Effect, MOB) validated by certified diabetologist Prof. Dr. Klaus Parhofer (LMU Munich). (Clinical Utility: 8.7/10).

EXIST Phase I Team Expansion: Clinical Research Coordinator (50%), Full-Stack Software Developer (50%) and Co-Founder for Business/DiGA Strategy (recruiting)

Advisory Board: Prof. Dr. Katarina Braune (Charité/HPI) – clinical trial co-PI; Prof. Dr. Klaus Parhofer (LMU) – protocol advisor.

Partnerships: 3 clinical trial sites (Charité, LMU network); regulatory consultants (ForTra); API discussions with Abbott/Dexcom.

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